

ORIGINAL ARTICLE

A simple contrastive embedding framework for low-resource fake news detection

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Abstract

Low-resource fake news detection aims at discerning between true and false claims from low-resource languages with scarce benchmark datasets. In this resource-constrained scenario, fake news data collected from online hoax reporting system is inherently skewed because human fact checkers mainly sample claims that are more likely to be fake or false. Instead of training end-to-end classifier on the extremely imbalanced dataset, our study investigates a simple framework based on contrastive learning and stacking-based ensemble learning as an alternate fake news classification pipeline for Indonesian language. Our empirical result shows that by combining contrastive-based embedding model—Contrast-BERT and ensemble of multilayer perceptrons (MLPs) in inference stage, we improve the precision score in fake news classification up to 26.64%, while maintaining accuracy and recall scores of above 75%, given extreme class imbalance ratio 1:24. Contrast-BERT is also superior to its counterparts in unsupervised topic clustering and evidence retrieval by nearly twofold. Furthermore, we observe that contrastive-based model follows a similar performance trend in Indonesian clickbait benchmark dataset. Contrast-BERT is more accurate and precise at predicting samples than end-to-end BERT classifier by up to 47%, given training subset with extreme imbalance ratio $\geq 1:19$.

Keywords Fake news detection · Low-resource scenario · Contrastive learning · BERT embeddings · Extreme class imbalance

1 Introduction

Fake news detection is a natural language processing (NLP) task that aims to discern between legitimate and fabricated information within textual content, such as news articles or other types of text distributed via social media platforms. If *real* news refers to an accountable bit of information or a real event, *fake* news is a counterfeit that is typically operationalized based on its level of facticity and the author's immediate intention, such as: fabrication, propaganda, manipulation, satire, and parody [1]. Fake news creates global and national issues, not only because it circulates the discourse that confuses the readers [2], but also because polarized information can highly affect election results [3] and political tensions in a country [2, 4]. News is socially constructed, making it easy for news creators to use fabricated news as propaganda or an advertisement for polarized readers. In addition, for people without digital literacy and fact-checking ability, fake news can resemble the look and feel of real news, and vice versa, worsening the dissemination of misinformation and disinformation on social media platforms. Therefore, the ability of machine learning algorithms or NLP methods to automatically identify and



flag texts as real or fake regardless of whether both categories share similar tone, sentiment, or emotion is crucial to control the spread of misinformation and disinformation and promote the dissemination of factual information instead.

However, obtaining large, accurately labeled data for training fake news detection models is costly and challenging. The reason is that high-quality labels are obtained based on the judgment of human experts to fact check claims or news articles with multiple external sources as references before deciding a verdict [5]. Generalizing the problem to an unseen target domain beyond English thus introduces language-specific challenges, particularly for training fake news detection models in a resource-scarce scenario [6]. In addition to the scarcity of linguistic resources and limited labeled data, the detection of fake news for low-resource language is challenging because there exist different cultural and linguistic nuances between English and non-English languages [7, 8]; domain specificity problem [9]; and inherited errors in translation-based augmentation approaches for low-resource language [10–12].

Although there has been increasing attention to developing methods for detecting fake news beyond English, such as for Arabic [13–15], Romanian [16], Malay [17], Hindi [18, 19], and Indonesian language [20], previous works have not addressed the problem in a realistic scenario of fake news detection with extreme class imbalance. In a realistic scenario, most of the real-world data for training fake news detection models comes from online public reports and government fact-checking initiatives, such as Masyarakat Anti Fitnah Indonesia (Mafindo) [21], which are mainly composed of false claims or fake news. In such a fact-checking platform, users submit claims and news articles that are more likely to be hoax or fake, resulting in data with an extremely imbalanced class representation where hoax becomes the majority class. Training machine learning models on such extremely imbalanced data can result in severe performance degradation, particularly on unseen data, where the distribution is different with the data on which the models are trained.

Our study aims at accurately discerning between true and false claims given extremely imbalanced real-world fake news data from Indonesian hoax reporting system. In particular, we want to approach the binary classification problem through the lens of contextual embeddings, moving beyond a standard end-to-end classifier. We hypothesize that by recasting the end-to-end fake news classification task into the objective of contrastive learning [22, 23], our approach allows learning data representation nearly unsupervisedly, thus mitigating bias or overfitting issues due to extremely imbalanced training samples. Contrastive learning uses a semantic paradigm in which explicit label supervision is substituted for semantic descriptors, such as positive and negative sentence pairs. In this study, positive and negative semantic descriptors are specifically described based on a collection of sentence pairs from Hoax and Non-Hoax class categories. Then, the model is enforced to recognize similar and dissimilar samples, given positive and negative data pairs. In inference, the trained contrastive model is mainly used as an embedding model for unseen evaluation data. The resulting text embeddings are then used as: (i) features for off-the-shelf MLP classifiers in a fake news classification pipeline; and (ii) features for clustering and retrieval evaluation tasks.

This paper first discusses two main categories of automated fake news detection approaches: (i) content-based; and (ii) context-based; and how they are adopted to languages beyond English (Section 2.1). In particular, we highlight the challenges for processing imbalanced fake news data in Indonesian, Bangla, and Czech; and how our data differ from currently available fake news and fact-checking datasets, as shown in Table 1 (Section 2.2). We also briefly discuss the comparison of methods for automated fake news detection, as shown in Table 2 (Section 2.3). Section 3 delves into research questions that guide the empirical experiments in this study. Section 4 explains the experimental setup and the resources used in this study. Finally, section 5 presents our empirical findings based on the performance measure in classification task and clustering; error analysis; and the ablation study on domain transfer, representation bias, and imbalance ratio. Our empirical experiments yield the following key insights:

- Contrastive learning captures fine-grained topics in fake news data as training signals.

Models based on contrastive learning—Contrast-BERT are observed to be more expressive than non-

Table 1 Comparison of currently available fake news and fact-checking datasets

Dataset	Natural	Domain	Language	#Claim	Low resource	Imbalance ratio		Evidence		
						Extreme imbalance	Ratio (%) Fake/ Real	Source	Retrieved	Annotated
English datasets										
FEVER [39]	✗	Multiple	English	109810	✗	✗	27:73	Wiki	✓	✓
Liar [40]	✓	Multiple	English	10269	✗	✗	64:46	Fact-checking	✗	✗
Snopes [41]	✓	Multiple	English	4236	✗	✗	73:27	Fact-checking	✗	✗
CHEF [42]	✓	Multiple	Chinese	7226	✗	✗	60:40	Internet	✓	✓
Low-resource fake news data										
Tallip-ID [6]	✗	Multiple	Indonesian	1960	✓	✗	50:50	✗	✗	✗
Clickbait-ID [43]	✓	✗	Indonesian	14928	✓	✗	42:58	✗	✗	✗
BanFakeNews [44]	✓	Multiple	Bangla, Bangladesh	50074	✓	✓	2:98	✗	✗	✗
CsFEVER [45]	✗	✗	Czech	124672	✓	✗	27:73	Fact-checking	✓	✓
CsFEVER-NLI [45]	✗	✗	Czech	218385	✓	✗	29:71	Fact-checking	✓	✓
IndoHoax (Ours)	✓	Multiple	Indonesian	12532	✓	✓	96:4	Fact-checking	✓	✓

English datasets are not covered in this study. Number of claims (#Claim) is based on training set, except for Tallip-ID where we view data as target test set only

contrastive baselines, particularly for capturing fine-grained topics inferred in training data, as shown in Tables 7 and 8. This capability is also transferable to new unseen domain within the same language category.

- Contrastive learning encourages fair representations, particularly toward implicit tone in fake news data.

We observe that Contrast-BERT is less biased than E2E BERT classifier when predicting Hoax samples under neutral and sensational tone categories as protected or sensitive attributes, as shown in Table 9.

Our contributions can be summarized as follows:

- We present a simple framework based on contrastive learning and stacking-based ensemble learning for low-resource fake news detection.

We demonstrate that Contrast-BERT consistently has higher precision scores than E2E BERT classifier in low-resource fake news classification task with an extreme class imbalance issue; particularly for Indonesian, Bangla, and Czech language.

- We are the first to provide comprehensive training and evaluation checklist (Section 3) for low-resource fake news detection through the lens of contrastive learning, moving beyond a standard end-to-end (E2E) classification objective.

Table 2 Comparison of currently available methods for automatic content-based fake news detection. We view OpenAI large language model (LLM) as a black-box method; thus, the only representation can be accessed is “Natural” representation of inputs and outputs

Method	#Params	Input feature		Representation				Contrastive learning	Backbone	Supervision	Research challenge		
		Claim	Evidence	Embedding			Natural				Low resource	Extreme imbalance	
				Word	Doc.	Graph							
Evidence-aware													
MAC	9.0M	✓	✓	✓	✓	✗	✗	✗	BiDAF/ LSTM	supervised	✗	✗	
GET	9.4M	✓	✓	✓	✓	✓	✗	✗	GNN	supervised	✗	✗	
GETRAL	9.4M	✓	✓	✓	✓	✓	✗	✓	GNN	supervised	✗	✗	
ReRead	249.5M	✓	✓	✗	✓	✗	✗	✗	BERT	supervised	✗	✗	
OpenAI LLM													
GPT-3.5	≈ 175B												
- Claim-only		✓	✗	✗	✗	✗	✓	✗	LLM	unsupervised	✗	✗	
- With evidence		✓	✓	✗	✗	✗	✓	✗	LLM	unsupervised	✗	✗	
GPT-4	≈ 1.8T												
- Claim-only		✓	✗	✗	✗	✗	✓	✗	LLM	unsupervised	✗	✗	
- With evidence		✓	✓	✗	✗	✗	✓	✗	LLM	unsupervised	✗	✗	
E2E BERT Classifier													
BERT	109.5M	✓	✓	✗	✓	✗	✗	✗	BERT	supervised	✗	✗	
BERT _(ensemble)	109.5M	✓	✓	✗	✓	✗	✗	✗	BERT	supervised	✗	✗	
BERT Embeddings													
No-Fine-tuning	109.5M	✓	✓	✗	✓	✗	✗	✗	BERT	unsupervised	✗	✗	
MLM Tuning	109.5M	✓	✓	✗	✓	✗	✗	✗	BERT	unsupervised	✗	✗	
Classif. Tuning	109.5M	✓	✓	✗	✓	✗	✗	✗	BERT	supervised	✗	✗	
Contrast-BERT (Ours)													
Pairwise	109.5M	✓	✓	✗	✓	✗	✗	✓	BERT	semi-supervised	✓	✓	
Triplet	109.5M	✓	✓	✗	✓	✗	✗	✓	BERT	semi-supervised	✓	✓	
Unsupervised	109.5M	✓	✓	✗	✓	✗	✗	✓	BERT	unsupervised	✓	✓	

2 Related work

2.1 Fake news detection in low-resource language

Automated fake news detection in low-resource language, which aims to identify fake news beyond the English language [15–18, 20], is a nontrivial problem due to the nature of the task that is mainly restricted by limited high-quality data and linguistic resources [6]. Previous studies have shown that data synthesizing by translation methods often produces low-quality data because there are cultural gaps and different linguistic nuances between English and non-English languages [7, 8]; and translation-based augmentation errors [12]. When a machine learning model is trained on these low-quality data, the performance of the model is expected to deteriorate [10].

In Indonesia, automated fake news detection research can be broadly divided into (i) *content-based* approaches—that mainly depends on linguistic features and dataset specificity; and (ii) *context-based* approaches, which consider multiple contextual information such as source credibility, temporal information, social engagement, network analysis, and multimodal analysis. Previous studies on content-based fake news detection use traditional machine learning methods such as Naïve Bayes [24] and XGBoost [25]; or BERT pretrained models [26] as end-to-end fake news classifier. However, none of these works have addressed problems in a realistic scenario of fake news detection with extreme class imbalance. Previous studies on the content-based classification of real and fake news also explore the role of subjective sentiment or tone in news data [27–31]. However, in a real-world fake news classification task, non-neutral or sensational tone can be found in both true and false claims. A shared-stylistic aspect across label categories introduces a new challenge to the content-based and sentiment-guided classifiers. On the other hand, to build an integrated fact-checking and fake news detection framework, context-based approaches [32–36] typically require high-quality social media data and relevant external knowledge, which are often inaccessible. It is also difficult to adopt context-based approaches in a new use case or new domain because the characteristics of data and experimental flows are specific to the authors’ own domain [33, 37, 38].

In this study, we further investigate whether currently available automatic fake news detection methods are able to address challenges in low-resource fake news detection task for Indonesian, Bangla [44], and Czech [45] language. Table 1 shows the low-resource fake news datasets in this study and how they differ from English datasets. Among low-resource fake news datasets, our Indonesian dataset is the only one with characteristics: (i) “*natural*”—data is not synthetically generated based on automatic machine translation methods, it is based on real-world claim report; (ii) “*extremely imbalanced*”; and (iii) contains retrieved evidences. BanFakeNews [44] has the closest characteristics to our data, particularly for containing an extreme class imbalance. However, the majority of samples in BanFakeNews come from authentic news sources (98%), compared to our data that are composed of 96% false claims; BanFakeNews also does not contain evidence to fact check the claims.

2.2 Class imbalance

The class imbalance problem, a condition in which one particular class significantly outweighs the other class, is a common challenge in real-world fake news detection. When a classifier or a model based on supervised learning is trained on this representation of bias, the model can carry and transfer such bias in unseen target data and the fake news domain [33, 46], introducing higher false positives where samples from the minority class (real news) are voted as fake news. To address class imbalance in fake news detection, researchers have proposed various methods that we broadly divided into two: (i) single-stage training; and (ii) dual-stage training. A single-stage training method focuses on improving model training from the news data have asymmetric to better sample data during the training stage. For example, latent space resampling [46] resamples class-biased latent vectors via oversampling, undersampling, and hybrid-sampling techniques. Methods based on ensemble learning [47–49] use multiple weak classifiers as independent experts, rather than relying solely on a biased classifier. Loss re-weighting [50] aims at increasing the influence of samples from the minority class category during the training stage. A dual-stage training method [51, 52], on the other hand, focuses on designing pretext tasks, such as predicting masked words [53] and contrasting samples [23] to generate training signals, instead of relying on explicit label supervision. The trained model is then used mainly as an encoder to project natural texts into multidimensional embedding space. In this study, we focus on a dual-stage training approach. We highlight the differences between our proposed approach and currently available baselines as shown in Table 2.

2.3 Contrastive learning

The core idea of contrastive learning [54–56] is to contrast between similar and dissimilar pairwise data points. In a generalizable definition of what is considered “similar” [57], a pair of data points is called a positive pair if both

points are different representations or views of the same data instance. Data points can be input-input (x_i, x_j) or input-label (x_i, y_j). For example, contrastive learning approaches in computer vision [22, 58, 59], learn from input-input pairs, i.e., image-to-image as pairwise data. In this setup, a positive pair is defined by multiple parallel augmentation of image input prior to training, such as by image rotation, random color cropping, random color distortions, and random Gaussian blur. However, contrastive learning in NLP, such as SimCSE [23], defines a positive pair as a pair of input and its contextual output representation (input-output) (x_i, y_c). In this setup, x_i is text input and y_c is a variation view of the said text. For example, x_i can be a text, y_c can be a summarized version of x_i [60–62]. A positive pair can also be defined as a pair of premises and its entailment hypothesis, such as contrastive learning based on Natural Language Inference (NLI) data [23].

Contrastive learning benefits our current study on fake news detection in low-resource language because it allows the model to learn representations in a self-supervised manner by comparing pairwise data, eliminating the requirement for learning from explicit label supervision. In our particular fake news domain where labeled data is scarce or costly to acquire, this training concept is a useful alternate objective as compared to directly learning from extremely imbalanced class representation that can may bring negative impact on unseen target domain or test data. Moreover, many previous empirical studies [63–67] demonstrate the ability of contrastive learning-based embedding models to produce robust representations in classification tasks, including the detection of fake news [51, 68–70].

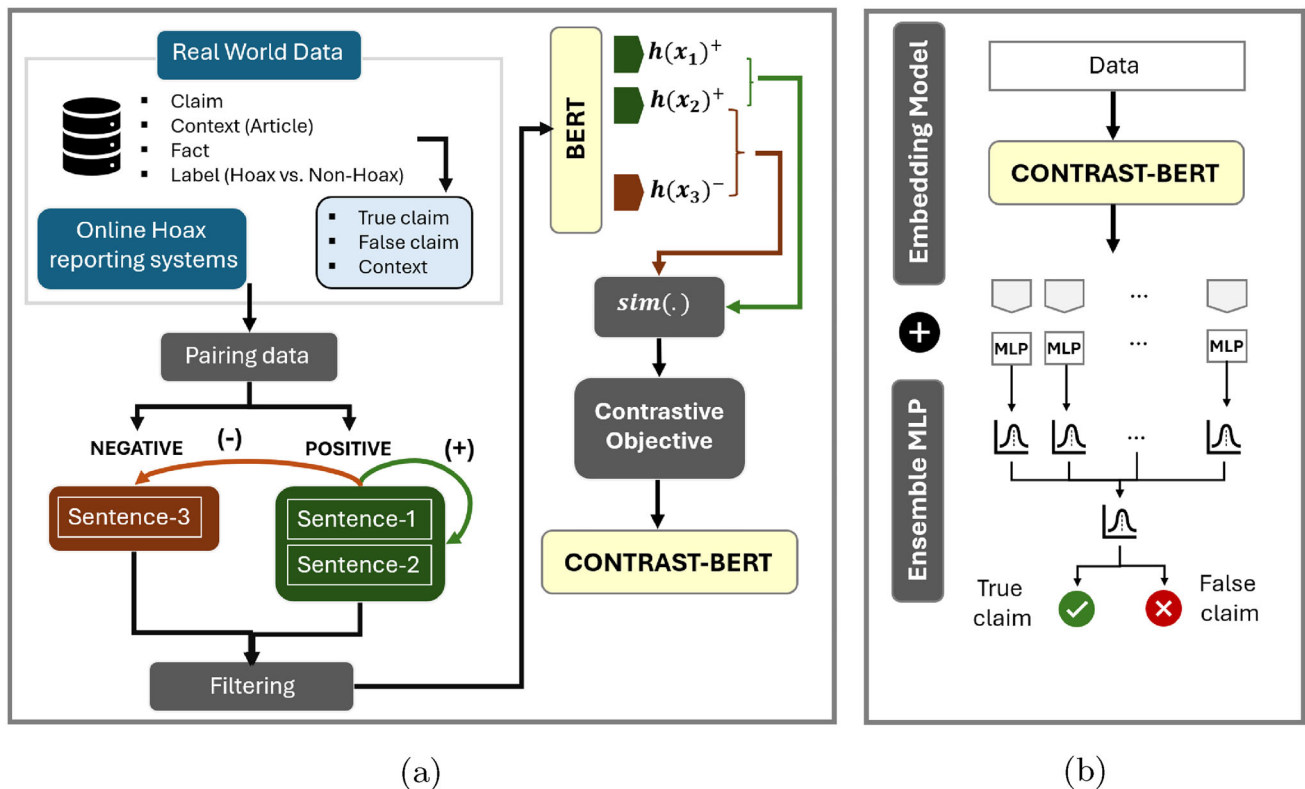


Fig. 1 Overview of our contrastive embedding framework for low-resource fake news detection. **a** Data preparation and training stage for contrastive-based BERT (Contrast-BERT). **b** Inference stage where Contrast-BERT is used as embedding model to project text into embedding or vector representation for ensemble MLP classifiers.

3 Methodology

Figure 1 illustrates an overview of our contrastive embedding framework for low-resource fake news detection. We collect data from hoax reporting systems (Table 4) where claims are mainly sampled from hoax candidates as the majority class, resulting in data with a severe class imbalance ratio of 1:24. Given {Claim (Title), Context (Article), Fact, Label}, we further preprocess the data to create positive and negative sentence pairs from both Hoax (false claim) and Non-Hoax (true claim) as input features for a contrastive objective (Section 3.1). In the data preprocessing stage, we incorporate filtering methods to exclude noisy sentence pairs (Section 3.2). Our fake news classification pipeline is described as sequential processes in the inference stage (Fig. 1b) where the trained contrastive-based model—**Contrast-BERT** is used as an embedding model to project text samples as vector representation as input features for independent MLP classifiers. Then, ensemble learning is described as training multiple MLPs in parallel as weak classifiers for smaller subsets with balanced class representation (Section 3.3). As a checklist for ablation studies and empirical evaluation, we outline our learning framework with the corresponding research questions as follows:

3.1 Contrastive objective

RQ1: *How to recast a fake news detection into a contrastive learning objective?*

A contrastive learning framework can be broadly divided based on its learning objective: (i) Unsupervised—where the model is trained on a collection of unlabeled sentences in a corpus; (ii) weakly supervised learning based on positive pairs—where positive pairs are explicitly defined to learn representations from “similar” samples as opposed to in-batch negative samples; (iii) supervised learning based on triplet data—where similar and dissimilar input features are explicitly defined as positive pairs and hard negatives respectively.

We adopt SimCSE contrastive learning framework [23] that originates from SimCLR [22], which approximates InfoNCE contrastive loss as a cross-entropy objective with in-batch negatives [71, 72]. If Hard Negative pair (x, x^-) is not explicitly defined, the negative samples are defined from in-batch training examples. If Positive Pair (x, x^+) is undefined, the positive sentence x^+ is defined based on implicit BERT augmentation of individual sentence. Contrastive learning objective with a batch of N sentence pairs is thus defined as:

$$\mathcal{L} = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^N e^{\text{sim}(h_i, h_j^+)/\tau}}, \quad (1)$$

where τ is a temperature hyperparameter and $\text{sim}(h_1, h_2)$ is cosine similarity function between sentences $\frac{h_1^T h_2}{\|h_1\| \cdot \|h_2\|}$.

3.1.1 Unsupervised

We construct three independent subsets for training and evaluating an unsupervised model: (i) title-only dataset $d^{\text{title}} = \{s_o^{\text{title}}, \dots, s_i^{\text{title}}\}$ from the title part of the claims for training Unsupervised-T; (ii) the concatenation between title and content $d^{\text{title+content}}$ for training Unsupervised-TC; and (iii) the concatenation between title and fact $d^{\text{title+fact}}$ for training Unsupervised-TF. From the resulting collection of sentences $\{x_i\}_{i=1}^m, x \in d$, a positive pair of sentences is composed of $\{x_i, x_i^+\}$ where x_i^+ can be defined as a nonessential variation of x_i without modifying its semantic meaning so that $x_i^+ = f(x_i)$. We utilize dropout masks on fully connected layers and attention probabilities (default $p=0.1$) in a standard training of Transformers and BERT-based models as $f(\cdot)$ —a noise function representing a minimal form of data augmentation, following a similar configuration in the previous study [23]. We simply project sentences x_i into the BERT encoder twice to obtain two independent

sentence embeddings with different dropout masks $\{z, z'\}$, representing positive pairs in Eq. 1. Thus, in an unsupervised setup, the objective becomes:

$$\mathcal{L}_{\text{unsup}} = -\log \frac{e^{\text{sim}(h_i^{z_i}, h_i^{z'_i})/\tau}}{\sum_{j=1}^N e^{\text{sim}(h_i^{z_i}, h_j^{z'_j})/\tau}}. \quad (2)$$

3.1.2 Weakly supervised (pairwise)

We construct pairwise data $\{q, s\} = \{(q_0, s_0^0), (q_0, s_0^1), \dots, (q_i, s_i^j)\}$ representing positive pairs (x_i, x_i^+) in this study by pairing the title as the query q_i and their target sentence s_i^j from document d_i , $d_i = \{s_i^0, \dots, s_i^j\}$ is composed of j sentences. The target document d_i is extracted from the content of the claim or reference to facts if available. The training objective for pairwise contrastive learning follows Eq. 1, such that $\mathcal{L}_{\text{pairwise}} = \mathcal{L}$.

3.1.3 Supervised (triplet)

We extend the pairwise setup (x_i, x_i^+) to triplets containing a negative sample for each positive pair (x_i, x_i^+, x_i^-) . For each positive pair (x_i, x_i^+) from a particular class category, **Hard Negative** x_i^- is explicitly defined by randomly sample titles from the opposite class. The training objective is thus defined by:

$$\mathcal{L}_{\text{triplets}} = -\log \frac{e^{\text{sim}(h_i, h_i^+)/\tau}}{\sum_{j=1}^N (e^{\text{sim}(h_i, h_j^+)/\tau} + e^{\text{sim}(h_i, h_j^-)/\tau})}. \quad (3)$$

3.2 Filtering

RQ2: *How to automatically clean pairwise data to ensure an effective and efficient contrastive learning?*

Not all the sentences in the corresponding article or the reference to the fact are equally relevant to the given claim. This problem is explicitly defined as a noise removal task in [73] where noisy data is removed using an independent BERT classifier. [51] convert claims and evidence to graphs to capture the long-distance dependency of relevant information. In fake news detection task, **noisy** sentence pairs can be present due to topic overlap, stylistic similarity, and shared entities across label categories (real news vs. fake news) [74–76], exemplified in the following sentence pairs.

- False positive pairs due to topic overlap.

Example:

Claim A (fake news): “NASA confirms alien life on Mars.”

Claim B (real news): “NASA’s Curiosity rover has found iron-rich carbonates on Mars.”

Issue: Both claims discuss a topic about Mars, but they come from different label category. Treating the samples as positive pairs can propagate the classification errors because the model focuses more on topic semantic similarity, instead of fact-checking property as an important step in a fake news detection task.

- False positive pairs due to stylistic similarity.

Example:

Claim A (fake news): “Jokowi Draws Red Line: Malaysia Dared to Send Weapons to Australia!”

Claim B (real news): “Record-Breaking! Jokowi claims 187 Trillion Rupiah Poured Into Villages!”

Issue: Both claims share a sensational tone that can lead to error propagation because during the training stage the model focuses more on a stylistic element rather than the news content or context to be verified.

- False positive pairs due to shared entities.

Example:

Sample A (fake news): “Jokowi warns Malaysia against sending weapons to Australia.”

Sample B (real news): “Jokowi claims to have disbursed 187 trillion rupiah in village funds.”

Issue: Both claims discuss the same entities (Name entity: People). Using this sentence pair as a training sample can lead to a model that focuses only on associating entity-related news as factor to classify sentences under the same label category.

To mitigate the presence of noises when pairing sentences, this study adopts two filtering methods used in previous works: (i) Consistency filtering [77, 78]; and (ii) Threshold filtering [77]. In particular, we focus on the following aspects when deciding whether a sampled pair can be included in training and evaluation dataset.

1. The relevance between the claim and the corresponding news content and evidence.

The concept of sentence pairing can be associated with the claim verification stage in fact-checking. That is, given a claim and the corresponding content and evidence or factual knowledge, the task is to verify the factuality of the claim relevant to the news content and evidence. In a content-based fake news detection and a contrastive learning approach, this verification task can be translated to a pairing task between claim and content or evidence. First, we split the news contents and evidence into sentences. Then, each claim is paired with the list of sentences from the news content and evidence to verify the claim. The resulting sentence pairs are then filtered using consistency filtering between claim as a *query* and sentences included in top-*k* retrieval.

Consistency filtering [77, 78]

We project query *q* and target sentences *s* onto pretrained embedding models to produce query and corpus embeddings. For each query, we find top-*k* (*k* = 5) most similar sentences in the corpus to the query through cosine similarity. Using nine independent pretraining embedding models listed in Table 3, we filter out samples or sentence pairs that are not in the top*k* retrieved sentences by one of the pretraining embedding models.

2. The threshold of triplet similarity between positive and negative pairs.

To prevent a false negative pair problem [74], we incorporate a threshold measure to distinguish between positive and negative pairs. This filtering method is specific to triplet data pairs that we constructed by randomly sampling negative sentences from the opposite label category.

Threshold filtering [77]

Table 3 Publicly available base pretrained models for BERT in this study based on model versioning in HuggingFace hub repository

Category	Pretraining	Base model
Symmetric semantic similarity	Indonesian	(1) LazarusNLP/simcse-indobert-base
	Indonesian	(2) LazarusNLP/simcse-indoroberta-base
	Indonesian	(3) LazarusNLP/congen-indobert-base
	Indonesian	(4) LazarusNLP/congen-simcse-indobert-base
	Multilingual	(5) sentence-transformers/paraphrase-multilingual-mpnet-base-v2
Asymmetric semantic search	English	(6) sentence-transformers/msmarco-roberta-base-v3
	Indonesian	(7) LazarusNLP/s-indobert-base-mmarco
Classification-based Pretraining	Indonesian	(8) indolem/indoberttweet-base-uncased
	Indonesian	(9) cahya/distilbert-base-indonesian
	Indonesian	(10) YagiASAFAS/indonesia-news-classification-bert

A threshold k is defined as the similarity difference between positive pairs and negative pairs $r(q, p) - r(q, n) > k$, so that triplets that have $k < 0.2$ are filtered out. The intuition of this filtering method is to ensure that positive pair has higher similarity score than negative pair.

3.3 Class imbalance

RQ3: *How to address extreme class imbalance problem in current fake news data?*

We define two sequential stages for reducing the overfit and the bias of the model toward the majority class.

Contrastive learning Contrastive learning as our main training framework in this study, allows BERT-based models to extract meaningful representations from unlabeled data by leveraging the concept of pairwise similarity and dissimilarity as training signals. Its advantages outnumber the end-to-end classifiers with the resampling method. It can learn useful features even when data is scarce and imbalanced [79]. On the other hand, resampling methods are known with numerous limitations [80], such as: (i) information loss because random undersampling discards data from the majority class so it potentially removes informative or useful samples; (ii) overfitting because random oversampling introduces data redundancy.

Ensemble MLPs We employ stacking-based ensemble learning [49] to train multiple weak off-the-shelf MLP classifiers, given the resulting embeddings from all embedding models in this study. Concretely, we divide training data into ten (10) subsets where each subset contains balanced representations from both class labels (Hoax and Non-Hoax) and four data sources, resulting in 500 Hoax samples and 500 Non-Hoax samples for each data subset. We train an independent MLP classifier for each training subset, resulting in 10 weak MLP classifiers. During the inference stage, we average the probability outputs of all ten weak classifiers, resulting in the final prediction. The ensemble method combines multiple weak classifiers to create a stronger and more robust classifier. Compared to the resampling method, ensembles can better mitigate bias due to class imbalance without discarding potentially informative samples and duplicating samples in training data.

3.4 Pretraining choice

RQ4: *Does pretraining choice affect model performance on fake news classification?*

To further investigate the impact of pretraining objective to build a contrastive learning framework for fake news detection, we classify ten (10) pretrained embedding models based on three broad categories: (i) Symmetric semantic similarity—the pretrained embedding models are trained to measure similarity between short texts (sentences); (ii) Asymmetric semantic search—the pretrained models are trained in a semantic-based retrieval task where query and document pairs have a different text length; and (iii) classification-based pretraining—the pretrained models are fine-tuned or distilled on multiclass classification task. The intuition is that fake news data have asymmetric characteristics where claims and articles or facts are in different lengths, similar to data characteristic in pretraining objective based on asymmetric semantic search. So, we want to compare whether such data assumption holds based on how the fine-tuned model performs on fake news classification as target domain. We list the pretrained models and their corresponding category in Table 3.

3.5 Evaluation tasks

RQ5: *How to evaluate contrastive learning in fake news detection domain?*

We define three main evaluation tasks for the evaluation of end-to-end classifiers and embedding models in this study as follows:

3.5.1 Fake news classification

We evaluate the models in the binary classification task based on two performance measurement settings: (i) Embedding model + Ensemble MLP classifiers; (ii) end-to-end classifier (BERT for sequence classification). In (i), test samples are projected onto the trained embedding models, including the proposed models based on contrastive learning. The resulting embedding is used as input features for ensemble MLP classifiers. In (ii), test samples are used as input for the trained BERT classifier, a baseline model with fully supervised learning, resulting in final class probability output.

3.5.2 Clustering

We use clustering evaluation to inspect the quality of embeddings based on their proximity distance in a presumably optimal embedding space. That is, whether the embeddings are naturally grouped into compact clusters such that similar text entities are being close to each other, while dissimilar texts or neighboring clusters of embeddings are sufficiently far from each other. In general, the clustering evaluation in this study serves as a compliance measure to objectively inspect whether the proposed contrastive embedding method conforms to two key properties for contrastive learning. Alignment and uniformity [81]. It is important to note that a clustering method is not a classification tool such that compact clusters are not necessarily faithful to the predefined class or label category. Our premise is that supervised embeddings may represent clusters of a fake news classification task ($k=2$) because the embedding model has been trained to fit the predefined binary class representation of Hoax and Non-Hoax samples. On the other hand, unsupervised and contrastive embeddings learned text representation almost unsupervisedly, thus the embeddings may only capture linguistic entities found in the given textual content, such as topics. To be fair, we evaluated the quality of the embeddings based on two clustering setups: (i) $k=2$; (ii) $k=40$. We manually create fine-grained topic labels for each sample in test data, resulting in 40 topic categories as ground-truth labels for evaluating clustering performance. The list of annotated topics will be provided in our publicly shared data and code repository.

3.5.3 Retrieval

In an integrated automated fact-checking framework, a retrieval task is included in the fact or evidence retrieval stage, and the final verification of the claim is included as a filtering method or noise removal task. Therefore, it is important to inspect the embedding model capability as part of a fact-checking system that focuses more on content-based detection task. In this evaluation task, give the reported claim as a query, the objective is to retrieve the relevant information that can either support or refute the claims from article and fact references metadata in corpus.

4 Experiments

4.1 Low-resource datasets

We compare the performance results between our proposed approach and the baselines on non-English language resources¹:

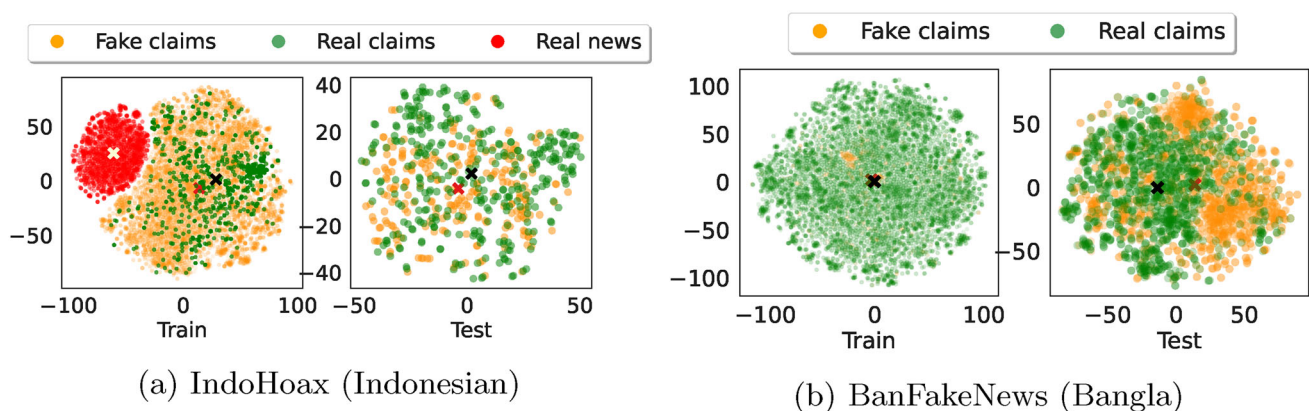
- IndoHoax—our constructed Indonesian fake news dataset, which is extracted from the real-world fact-checking and hoax reporting system in Indonesia² from the timeline of 2018–2023, resulting in training and

¹ All datasets are available at <https://anonymous.4open.science/r/contrast-BERT>.

² <https://mafindo.or.id/>.

Table 4 Real-world fake news data based on Indonesian online hoax reporting systems. The average #token is based on LazarusNLP/simcse-indobert-base tokenizer on training data. C0=“Non-Hoax,” while C1=“Hoax”

Source	Sample Size		Class Proportion (%)				Average #Tokens			Fact (%)	
	Train	Test	Train		Test		Title	Content	Fact	Train	Test
			C0	C1	C0	C1					
Mafindo	6983	199	1.2	98.8	50	50	14 ± 5	94 ± 114	227 ± 175	98.3	97.5
Saber-Hoax Jabar	4523	198	3.5	96.5	50	50	12 ± 3	52 ± 37	148 ± 95	89.7	72.5
Opendata Jabar	354	99	15.5	84.5	50	50	14 ± 4	N/A	N/A	0.0	0.0
Klinik-Hoax Jatim	492	98	100	0.0	100	0.0	12 ± 3	47 ± 25	103 ± 41	29.3	0.0
Total	12352	594	6.4	93.6	58.3	41.7	13 ± 4	78 ± 94	196 ± 154	89.6	56.7

**Fig. 2** Embedding visualization for Indonesian and Bangla fake news data.

test datasets as shown in Table 4. Unlike the other low-resource datasets in this study, our Indonesian dataset is composed of (i) diverse topics (i.e., *Politics, Law and Criminals, Social, Economy, Education, Technology, Sports, Health, Entertainment*); (ii) fine-grained categorization of misinformation and disinformation (i.e., *Fabricated content, Impostor content, Satire, Misleading content, Manipulated content, False context, False connection*), and extremely class imbalance ratio (96:4) between fake and real claims.

Preprocessing: We first clean IndoHoax data by removing explicit class identifier or sentences in news content and evidence that summarize whether a claim is true or false, for example: “This is Hoax,” “[FALSE CLAIM].” We remove new content and evidence that contain less than 20 words. For training a contrastive learning objective, we create sentence pairs by pairing claims and sentences in news content and evidence. We remove noisy sentence pairs both in pairwise and triplet datasets by filtering data using consistency and threshold filtering (Section 3.2).

Class Ratio: Our IndoHoax dataset is inherently imbalanced, where the ratio between fake news samples and real news is 96:4 (Table 4). In our preliminary experiment, we exercise the fake news classification task by adding more real news data from reputable news sources to the training dataset. However, because the characteristic of real news data from the newly added domain is different to the current real news subset, the classification performance does not improve, implying a *stylistic* problem inherently found in the current dataset. We later discuss this stylistic problem in Section 5 (Fig. 2). Therefore, in this study, we focus more on

¹ All datasets are available at <https://anonymous.4open.science/r/contrast-BERT>.

² <https://mafindo.or.id/>.

investigating methods for training a fake news classification task. We view a new approach with data augmentation strategy to enforce models to be invariant to topic overlap, stylistic similarity, and shared entities as a future research direction.

- Tallip-ID [6]—Indonesian-translated fake news data from multilingual fake news detection. The data contains a ground-truth topic categorization for each claim. We mainly use this data for zero-shot performance analysis on topic clustering;
- Clickbait-ID [43]—Indonesian clickbait benchmark dataset. Specific for clickbait data, we create multiple training subsets with different class imbalance ratio to further inspect the performance consistency across domain and imbalance level. This particular preprocessing step produces: (i) *balance* subset (50:50); (ii) *mildly balanced* (70:30), (75:25); (iii) *moderately imbalanced* (80:20), (85:15); and (iv) *extremely imbalanced* (90:10), (95:5), (98:2);
- BanFakeNews [44]—Dataset for clickbait detection in Bangla language. This dataset consists mainly of headlines and the corresponding content of news articles. Each headline is classified into either “clickbait” or “nonclickbait”;
- CsFEVER and CsFEVER-NLI [45]—Fact-checking datasets that are constructed by translating FEVER dataset from English to Czech language. We further preprocessed these datasets to produce subsets of data with a similar class imbalance ratio to the Indonesian dataset (IndoHoax).

DeClare Format

To be able to train evidence-aware fake news detection approaches on our Indonesian dataset, we reconstructed data into a new structure defined in DeClare [82]. The data transformation results in **10,755** training set and **337** test set for our Indonesian data. The new DeClare formatted dataset is mainly composed of samples containing claims, the corresponding single summarized evidence, and the URL source of evidences. We discarded samples that contain no evidence.

4.2 Baselines

Table 2 shows automatic fake news detection methods that are being investigated in this study. We divide baselines based on three categories:

1. End-to-end (E2E) classification methods, which are further divided into:
 - (a) Evidence-aware approaches:
 - MAC (Multi-head Attention network for Fact-Checking) [83]—The method employs multi-head attention mechanisms to analyze and cross-reference multiple pieces of evidence against a claim;
 - GET (Graph-based sEmantic sTructure) [84]—The method models claims and evidence as graph-structured data to capture long-distance semantic dependencies through neighborhood propagation;
 - GETRAL (Graph-based sEmantic structure mining framework with ConTRActive Learning) [51]—The method integrates supervised contrastive learning and adversarial augmented instances to enhance GNN representation learning and reduce sensitivity to local evidence;
 - ReRead [73]—The method involves two key steps: (i) training an evidence retriever to obtain interpretable evidence based on faithfulness and plausibility; and (ii) training a claim verifier to reexamine the retrieved evidence to enhance accuracy.
 - (b) Content-based approach based on BERT pretrained language model [85]. Specific to BERT-based methods, we compare BERT models with and without fine-tuning strategy. For fine-tuning BERT, we mainly focus on a simple mask language modeling (MLM) fine-tuning objective [86] and a binary classification task on target fake news domain.

2. Two-stage learning [50]—To provide a fair evaluation, we use **BERT embeddings** without contrastive learning as a baseline model for two-stage learning approach.
3. Large language models (LLMs)—LLMs have been utilized as an AI agent to predict the veracity of claim for fact-checking and fake news detection task [73, 87]. We use OpenAI GPT-3.5-turbo (gpt-3.5-turbo-0125) and GPT-4-turbo as LLM baseline in this study. For evaluating LLM performance, we compare the prediction output when LLM use “claim” as the only input to the model and “claim + evidence” as input. The class probability score is computed based on the probability of LLM outputs tokens: “REFUTES” (*fake*) or “SUPPORTS” (*real*). Specific for LLM with evidence as auxiliary input, we do not output the token probability score because the model generates prediction with reasoning (Chain-of-Thought [88]).

4.3 Training

For all training and evaluation schemes, we employ NVIDIA DGX-A100 GPU with 40GB RAM and CUDA version: 12.2. The hyperparameters³ are configured independently for each model. To ensure the reproducibility aspect, we configure our evaluation code to run deterministically by using a same set of random seeds across all models in this study.

5 Result and discussion

5.1 Fake news classification

Performance on Indonesian fake news data We investigate the capability of the proposed Contrast-BERT for predicting whether a claim is considered *fake* or *real*. Table 5 shows the performance comparison on Indonesian fake news data. We observe that both Pairwise Contrast-BERT and Unsupervised Contrast-BERT outperform E2E BERT by maintaining a balance score between Precision (P) and Recall (R) when predicting fake and real claims as positive class label. However, fine-tuning BERT on extremely imbalanced data with a supervision does not yield a better outcome. Both masked language model (MLM) and classification-based tuning method fail to improve the performance of pretrained BERT on target domain, suggesting that utilizing a pretrained model based on contrastive learning (base model “LazarusNLP/simcse-indobert-base”) is adequate for the current fake news detection task. Overall, Contrast-BERT is less prone to the overconfidence errors, which is demonstrated by the lowest brier loss (15.8), as shown in Table 5.

Specific to evidence-aware approaches, we observe that a simple hierarchical attention networks can outperform approaches based on graph neural networks (GNNs), such as GET and GETRAL. We believe that a lower performance score of GNNs is mainly due to the networks are not able to capture useful structure inferred in IndoHoax. Compared to the English fake news and fact-checking datasets shown in Table 1, IndoHoax does not contain multiple evidences and the corresponding sources.

Lastly, utilizing a more powerful large language model, such as OpenAI GPT-4; and grounding the model with an evidence, boost the overall classification performance. However, GPT-4 comes at a higher price. Comparing GPT-4 ($\approx 1.8T$) and models with smaller parameter size, such as BERT (109.5M), thus, is rather an unfair performance comparison.

Analysis on dataset difficulty We investigate whether Indonesian dataset poses a different challenge than the other low-resource fake news datasets in this study. First, we project data into embedding space via OpenAI “text-embedding-3-large.” Based on the embedding visualization in Figure 2, specific to the Indonesian dataset (IndoHoax), we observe that simply adding more data from real news articles does not necessarily address

³ The hyperparameter configuration is available at <https://anonymous.4open.science/r/contrast-BERT>.

Table 5 Performance on our Indonesian fake news dataset (IndoHoax) based on claim and evidence as merged input features. Base model for BERT: LazarusNLP/simcse-indobert-base

Method	$\uparrow$ ACC	$\uparrow F_1^{\text{Macro}}$	$\uparrow F_1^{\text{Micro}}$	Real News			Fake News			$\downarrow$ Brier Loss
				$\uparrow F_1$	$\uparrow P$	$\uparrow R$	$\uparrow F_1$	$\uparrow P$	$\uparrow R$	
Evidence-aware										
MAC _(ensemble)	72.7*	71.1	72.7*	64.3	72.8	57.6**	77.8**	72.6**	83.9	19.2
GET _(ensemble)	63.5	52.2	63.5	28.9	86.2	17.4	75.4*	61.4	97.9	20.9
GETRAL _(ensemble)	66.2	57.1	66.2	37.4	89.5	23.6	76.8**	63.2	97.9	20.8
ReRead _(ensemble)	44.5	34.4	44.5	8.78	75.0	4.66	60.1	43.3	97.9	26.1
OpenAI LLM										
GPT-3.5-turbo										
- Claim-only	67.5	67.5	67.5	66.9	58.9	77.2**	68.1	78.2**	60.3	25.8
- With evidence	85.9**	84.7**	85.9**	80.4**	95.7	69.3**	89.1**	81.9**	97.8	N/A
GPT-4-turbo										
- Claim-only	72.7*	70.8	72.7*	63.2	74.5	54.9**	78.3**	71.9**	86.0	24.1
- With evidence	92.8**	92.4**	92.8**	90.8**	94.8	87.2**	94.1**	91.6**	96.7	N/A
E2E BERT Classifier										
BERT	68.7	68.2	68.7	64.2	95.4	48.4	72.1	57.5	96.8	31.0
BERT _(ensemble)	77.7**	77.7**	77.7**	78.4**	89.9	69.5**	77.1**	67.9**	89.1	17.9
BERT Embeddings										
No-Fine-tuning	75.2**	74.7**	75.2**	78.3**	79.6	77.1**	71.1	69.6**	72.6	16.3
MLM Tuning	74.6**	73.6**	74.6**	78.6**	76.8	80.6**	68.6	71.1**	66.3	18.2
Classif. Tuning	70.4	70.1	70.4	67.0	94.7	51.9	73.1	59.0	95.9	29.1
Contrast-BERT										
Pairwise	77.3**	76.8**	77.3**	80.2**	81.3	79.1**	73.3	72.1**	74.7	15.9
Triplet	74.7**	74.5**	74.7**	76.9**	81.9	72.5**	72.1	67.1**	77.9	16.9
Unsupervised	76.1**	75.5**	76.1**	79.3**	79.8	78.8**	71.7	71.1**	72.3	15.8

Significance value $p < 0.05$ (*) and $p < 0.01$ (**) are computed based on E2E BERT classifier with 95% confidence

the class imbalance problem. The reason is that real news articles come from a different distribution than “real” claim samples in IndoHoax training set. It is also interesting to observe that both fake and real claims in IndoHoax are highly overlapping, suggesting an inferred stylistic and linguistic challenge for content-based fake news detection methods.

Error analysis

We further investigate the prediction errors of classification models in this study, particularly Contrast-BERT (Pairwise) and end-to-end (E2E) BERT classifier as follows:

1. Statistical summary of errors (false positive and false negative)

Table 6 shows the statistical summary of prediction errors and the corresponding model’s confidence. In this study, false positive (FP) refers to the type of prediction error where the model predicts real news and true claim as **False** or “fake news,” while false negative (FN) is the opposite label misclassification.

Based on the comparison of model confidence in Table 6, we observe that the model based on contrastive learning (Contrast-BERT) is less confident than E2E BERT classifier and native BERT embedding when misclassifying samples. However, Contrast-BERT is overconfident for predicting true claim, resulting in a higher number of false negative. We further analyze misclassified samples based on both false positive and false negative error types based on topic, name entities, and stylistic or tone implied in test samples.

2. Qualitative findings

We inspect the prediction errors of the proposed Contrast-BERT based on topic overlap, shared name entities, and stylistic aspect of the news or claims, such as the presence of sensational tone. We observe that a high topic overlap between class labels is present in the current Indonesian dataset, with an exception that particular topic categories are dominantly present in one class. For example, “Government Policy” occurs more in true claims, while “Financial Scams,” “Political Figures,” and “Sensational/Viral Video” occur more in false claim samples. We observe that a higher proportion of false positives come from the three dominant topics in false claim samples (“Financial Scams,” “Political Figures,” and “Sensational/Viral Video”), suggesting that the model may use cues or signals belong to a fake news category, such as name entities and sensational tone commonly found in false claims.

- Correctly classified “true claims” (real news)

Topics: Campaign & Election (13), COVID-19 Outbreak (21), COVID-19 Vaccine Hoaxes (6), COVID-19 Disinformation (41), Fake Job Openings (4), Financial Scams (45), Food & Health Rumours (2), Government Policy (74), Healthcare (27), Natural Disaster (15), Policy & Public-Service Misinformation around COVID-19 (9), Political Figures (26), Sensational/Viral Video (20).

- Correctly classified “false claims” (fake news)

Topics: Campaign & Election (18), Counterfeit Account/Letter (14), COVID-19 Outbreak (22), COVID-19 Vaccine Hoaxes (3), COVID-19 Disinformation (54), Fake Job Openings (5), Financial Scams (67), Government Policy (9), Healthcare (40), Natural Disaster (4), Political Figures (29), Sensational/Viral Video (66).

- False positive

Topics: Campaign & Election (7), Counterfeit Account/Letter (2), COVID-19 Outbreak (6), COVID-19 Vaccine Hoaxes (2), COVID-19 Disinformation (7), Fake Job Openings (4), Financial Scams (23), Food & Health Rumours (2), Government Policy (3), Healthcare (10), Natural Disaster (5), Political Figures (14), Sensational/Viral Video (14).

Example:

Topic: Political Figures

Text:

Claim: Coordinating Minister Luhut Denies Sending an Open Letter to Amien Rais and Prabowo.

Fact: Coordinating Minister for Maritime Affairs, Luhut Binsar Pandjaitan, firmly stated that he never sent an open letter to Amien Rais and Prabowo Subianto, contrary to the news circulating recently.

Label: true claim

Predicted: false claim (fake news).

Topic: Sensational/Viral Video

Text:

Table 6 Summary of errors based on false positive and false negative for the proposed Contrast-BERT and E2E BERT classifier

Metric	Contrast-BERT Pairwise (ours)			E2E BERT			BERT + Ensemble MLP		
	TP	FP	FN	TP	FP	FN	TP	FP	FN
Mean confidence	0.837	0.713	0.728	0.996	0.994	0.991	0.996	0.989	0.958
Median confidence	0.878	0.693	0.731	1.0	1.0	1.0	1.0	1.0	1.0
Min confidence	0.502	0.500	0.504	0.590	0.687	0.929	0.528	0.555	0.584
Max confidence	1.0	0.982	0.993	1.0	1.0	1.0	1.0	1.0	1.0
Count	459	73	62	408	178	8	418	166	10
Total	459	135		408	186		418	176	
Inference Size	594								

Claim: A RARE SIGHT—SNOW FALLS IN SAUDI ARABIA.”

Fact: Snow has fallen in the southwestern mountainous regions of Saudi Arabia. Residents were seen joyfully enjoying the sight of deserts blanketed in white snow. According to Arab News on Wednesday (January 15, 2020), the areas that experienced snowfall include the northern parts of Tabuk, Jabal Al Lawz, Al-Daher, and the Alqan mountains. These snowy regions in Saudi Arabia are popular tourist destinations.

Label: true claim

Predicted: false claim (fake news).

- False negative

Topics: Campaign & Election (6), Counterfeit Account/Letter (6), COVID-19 Outbreak (6), COVID-19 Vaccine Hoaxes (1), COVID-19 Disinformation (11), Fake Job Openings (4), Financial Scams (15), Food & Health Rumours (2), Government Policy (3), Healthcare (7), Natural Disaster (2), Political Figures (5), Sensational/Viral Video (4).

Performance across languages

Figure 3 shows the performance comparison of four methods: (i) our proposed **Contrast-BERT**; (ii) E2E BERT for Sequence Classification; (iii) OpenAI GPT-3.5-turbo; and (iv) OpenAI GPT-4-turbo. In this experiment, we configure all datasets to be extremely imbalanced, resulting in imbalance ratio between fake news and real news as follows: IndoHoax (96:4); BanFakeNews (4:96); CsFEVER (96:4); and CsFEVER-NLI (91:9). In particular, we observe that the proposed Contrast-BERT outperforms E2E BERT when predicting “Real” claims, which are sampled from the minority class category, as shown in Figure 3a. Compared to the other low-resource datasets, both unnatural Czech fake news datasets (CsFEVER and CsFEVER-NLI) introduces challenges for BERT-based models to predict samples from both minority and majority classes, following a similar low performance reported in the original paper [45]. Nevertheless, our Contrast-BERT is able to predict the minority class samples in both CsFEVER and CsFEVER-NLI, compared to E2E BERT that cannot recall all real news samples (F_1 score for real news = 0.0). It is also interesting to observe that OpenAI GPT-4-turbo, which is the most powerful Black Box model in this study, underperforms on our Indonesian datasets compared to the other datasets. This suggests that our Indonesian dataset introduces the remaining challenge for future study on content-based low-resource fake news detection.

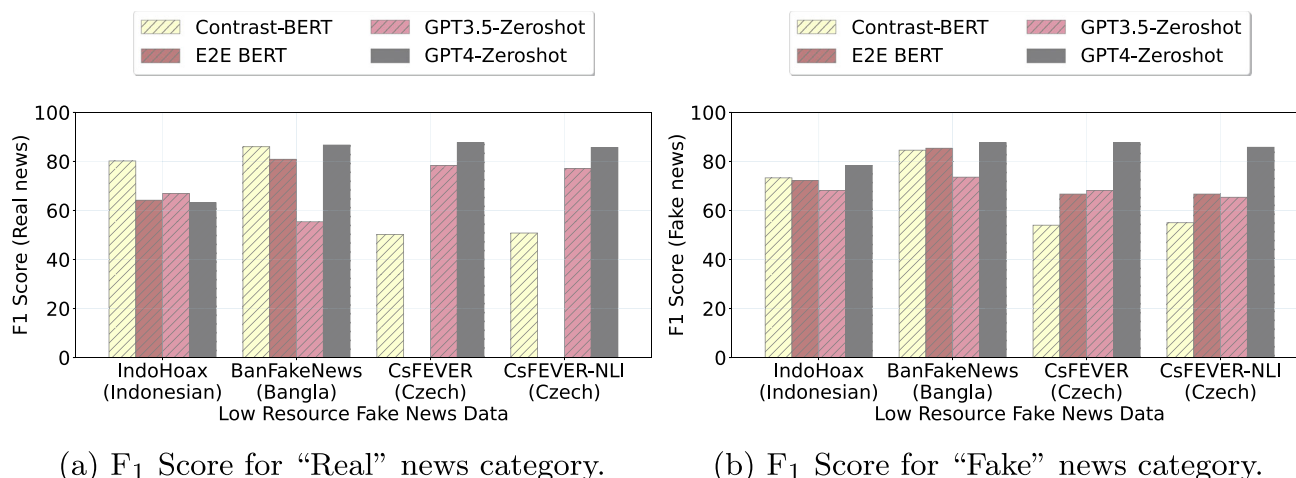


Fig. 3 Fake news classification performance (%) on low-resource fake news datasets based on F_1 Score for “Real” and “Fake” news.

Table 7 Performance on clustering and retrieval tasks. Base model: LazarusNLP/simcse-indobert-base

Method	Clustering								Retrieval	
	↑Completeness		↑Homogeneity		↑ARI		↑NMI		↑HR@5	↑MRR@5
	k = 2	k = 40	k = 2	k = 40	k = 2	k = 40	k = 2	k = 40		
Unsupervised Embeddings										
Zero-shot-TF	3.0e−4	0.332**	2.8e−4	0.365**	8.0e−4	0.084**	2.8e−4	0.347**	0.783**	0.716**
MLM Fine-tuning-TF	0.005	0.305**	0.005	0.339**	0.009	0.067**	0.005	0.321**	0.737**	0.642**
Supervised Embeddings										
E2E BERT-T	0.146	0.185	0.099	0.181	0.126	0.014	0.119	0.183	0.405	0.344
E2E BERT-TC	0.108	0.168	0.100	0.183	0.150	0.017	0.104	0.175	0.421	0.366
E2E BERT-TF	0.097	0.178	0.069	0.201	0.099	0.015	0.081	0.189	0.405	0.328
Contrast-BERT (ours)										
Pairwise-TCF	0.043	0.381**	0.043	0.415**	0.053	0.102**	0.043	0.398**	0.882**	0.835**
Triplet-TCF	0.009	0.437**	0.009	0.487**	0.006	0.154**	0.009	0.461**	0.862**	0.815**
Unsupervised-T	0.003	0.249**	0.003	0.262**	0.007	0.039**	0.003	0.256**	0.763**	0.696**
Unsupervised-TC	4.5e−4	0.242**	4.5e−4	0.261**	3.5e−4	0.027*	4.5e−4	0.251**	0.734**	0.656**
Unsupervised-TF	0.010	0.280**	0.010	0.310**	0.014	0.044**	0.010	0.295**	0.753**	0.685**

Significance value $p < 0.05$ (*) and $p < 0.01$ (**) are computed based on E2E BERT-T with 95% confidence

5.2 Clustering on Indonesian data

The clustering performance represents model's own inductive bias—the implicit structure that the model learns by connecting different entities in training data [89, 90]. Table 7 shows the performance comparison given the predefined number of clusters as $k=2$ and $k=40$. In binary clustering ($k = 2$), the number of clusters represents binary labels in fake news data, while in clustering with $k = 40$, the number of clusters represents fine-grained topics inferred in target data, which we manually define (Section 3.5.2). In general, **Contrast-BERT** models are observed to capture fine-grained topics than binary label representation, which is shown by higher completeness score (0.437), homogeneity (0.487), ARI (0.154), and NMI (0.461) in topic clustering experiment with $k = 40$. In contrast, supervised embeddings have higher scores when $k=2$. However, we observe that the natural clusters formed by the supervised embeddings do not represent the actual label supervision, which is shown by the considerably low overall performance score for $k = 2$ across multiple clustering evaluation metrics (< 0.2).

5.3 Retrieval on real-world data

Table 7 includes the performance scores for evidence retrieval task where the embedding models rank top-5 most relevant documents or evidences in the corpus given claim as a query. In general, we observe that both Pairwise and Triplet Contrast-BERT can retrieve at least 86.4% of relevant document references based on their top-5 prediction. In contrast, supervised embeddings have a poor retrieval performance ($\text{HR@5} \leq 0.5$ and $\text{MRR@5} < 0.4$). This result implies the capability of Contrast-BERT to be used in a retrieval-based fact verification pipeline, in addition to be used as embedding model in fake news classification pipeline.

5.4 Ablation study

We further investigate the impact of heuristic choices in our experimental design as follows:

Pretraining choice

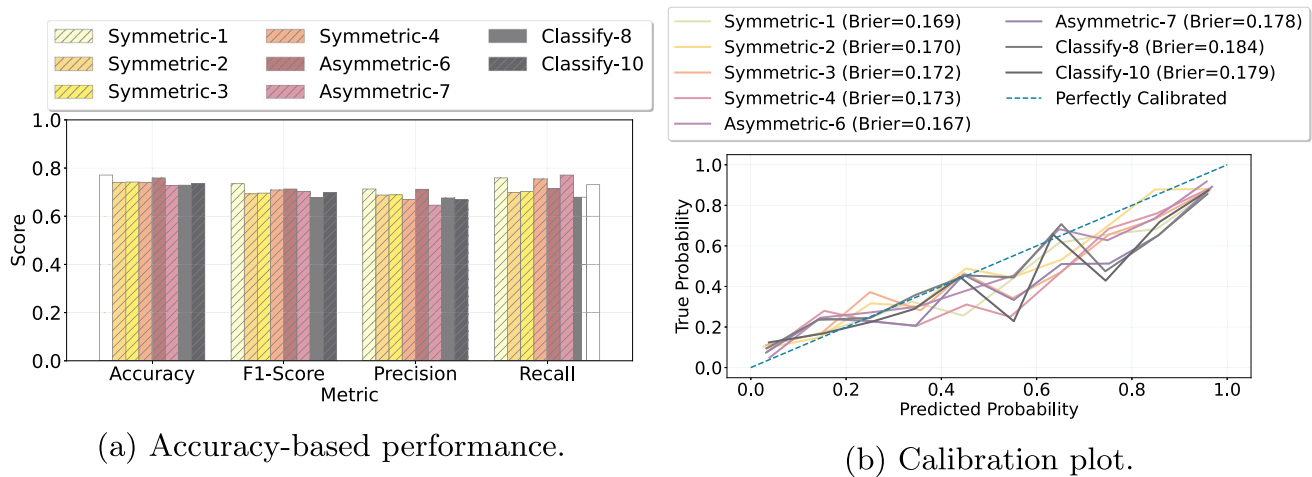


Fig. 4 Performance comparison based on choices of pretrained embeddings.

Figure 4 shows the impact of choosing pretrained embeddings based on our categorical constraints (Table 3). We observe that utilizing classification-based pretrained embeddings (Classify-8 and Classify-10) leads to performance deterioration, producing the lowest accuracy-based performance (Fig. 4a) and the highest brier loss (Fig. 4b). On the other hand, there is no such significant differences between utilizing pretrained embeddings in Symmetric and Asymmetric category, as shown by small performance difference based on Brier loss between Symmetric-1 and Asymmetric-6. In general, utilizing base models based on contrastive learning that has been pretrained on Indonesian Wikipedia pairwise data—**Symmetric-1** is shown to be beneficial in current fake news classification task.

Domain transfer

Table 8 Zero-shot clustering and retrieval performance on Indonesian-translated fake news dataset. Score is based on the average of 10 runs. Base model: LazarusNLP/simcse-indobert-base

Method	Clustering								Retrieval	
	↑Completeness		↑Homogeneity		↑ARI		↑NMI		↑HR@5	↑MRR@5
	k = 2	k = 7	k = 2	k = 7	k = 2	k = 7	k = 2	k = 7		
Unsupervised Embeddings										
Zero-shot-TF	5.5e−5	0.209**	5.5e−5	0.256**	−6.8e−4	0.132**	5.5e−5	0.230**	0.821**	0.669**
MLM Fine-tuning-TF	0.001	0.205**	0.001	0.233**	8.4e−4	0.154**	0.001	0.218**	0.636**	0.491**
Supervised Embeddings										
E2E BERT-T	0.013	0.056	0.002	0.058	0.001	0.048	0.003	0.057	0.438	0.328
E2E BERT-TC	0.006	0.064	0.003	0.072	0.005	0.100**	0.004	0.068	0.412	0.330
E2E BERT-TF	0.002	0.043	8.6e−4	0.042	−0.002	0.064	0.001	0.043	0.370	0.292
Contrast-BERT (ours)										
Pairwise-TCF	0.001	0.311**	0.001	0.377**	0.001	0.182**	0.001	0.341**	0.895**	0.738**
Triplet-TCF	1.3e−4	0.349**	1.2e−4	0.400**	−3.4e−4	0.274**	1.2e−4	0.373**	0.864**	0.715**
Unsupervised-T	0.002	0.162**	0.002	0.197**	0.002	0.105**	0.002	0.178**	0.710**	0.546**
Unsupervised-TC	0.006	0.180**	0.006	0.219**	0.006*	0.123**	0.006*	0.198**	0.726**	0.572**
Unsupervised-TF	0.002	0.167**	0.002	0.205**	0.002	0.119**	0.002	0.184**	0.743**	0.604**

Significance value $p < 0.05$ (*) and $p < 0.01$ (**) are computed based on E2E BERT-T with 95% confidence

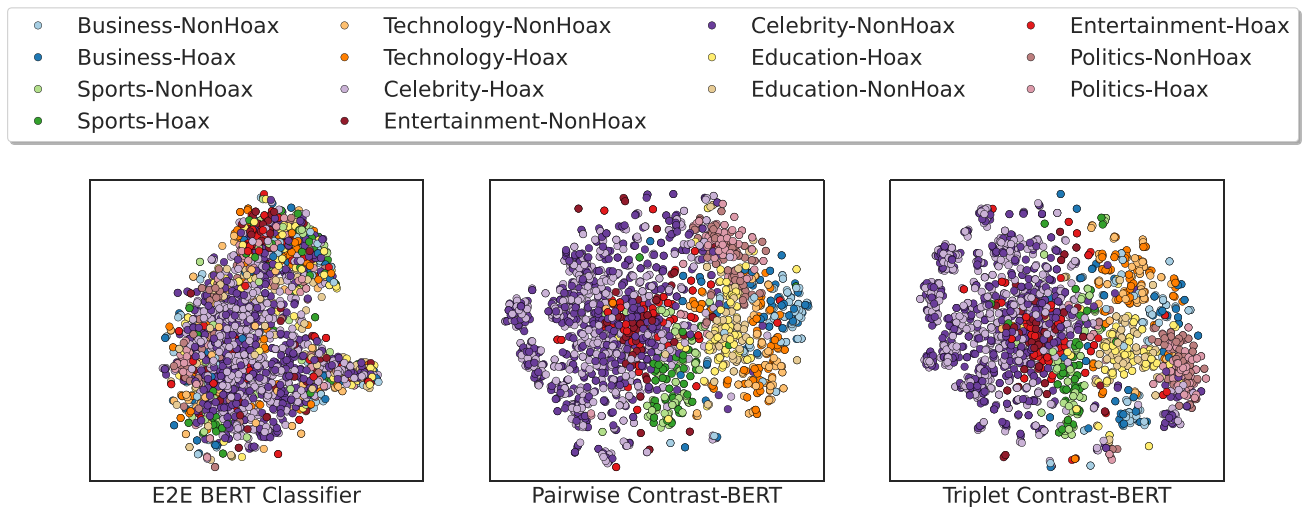


Fig. 5 2D Visualization of embeddings with TSNE, based on zero-shot embedding projection on unseen Indonesian-translated fake news data with seven topic categories

Table 9 Measuring bias of the embeddings on neutral tone (minor group category) vs. sensational tone (major group category) for “Hoax” label (Class label=1). Base model: LazarusNLP/simcse-indobert-base

Metric	Reference	E2E BERT-T (Title only)	E2E BERT-TF (Title+Fact)	Zero-shot	Pairwise Contrast-BERT
Statistical Parity	0	−0.3031	−0.2458	−0.0281	0.0240**
Disparate Impact	1	0.6421	0.6679	0.9279**	1.0448**
Four Fifths Rule	1	0.6421	0.6679	0.9279**	0.9571**
Cohen D	0	−0.7445	−0.5365	−0.0579	0.0483**
Equality of Opportunity Diff	0	−0.0038	0.0234	−0.0860	−0.0988
False Positive Rate Diff	0	−0.3035	−0.2301	−0.0119	0.0168**
Average Odds Diff	0	−0.1536	−0.1033	−0.0489	−0.0410**
Accuracy Diff	0	0.0478	0.0545	0.0464	−0.0553

Significance value $p < 0.05$ (*) and < 0.01 (**) are computed based on E2E BERT-T with 95% confidence

We investigate the zero-shot classification performance on unseen Indonesian-translated fake news dataset from multilingual fake news detection study [6]. This dataset contains explicit topic categorization (n topic = 7). Table 8 shows that Pairwise Contrast-BERT and Triplet Contrast-BERT consistently perform well on both topic clustering with $k=7$ and evidence retrieval task. The zero-shot performance results imply the generalization capability of models based on contrastive learning, particularly when unseen data contains sufficient contextual abstraction such as topic categories. Figure 5 shows 2D visualization of text embeddings from Indonesian-translated fake news data [6]. We observe that Contrast-BERT models are better at capturing topics as natural clusters of data.

Representation bias

We further inspect whether the resulting embeddings inherit biases toward textual tone of fake news data. Although most of Hoax data are created mainly using sensational tone, it is important to note that disregarding this characteristic of the data as a sensitive attribute or protected feature and allowing machine learning models to directly pick up the textual tone as essential features during the prediction stage is considered harmful from the perspective of fairness of AI [91, 92]. First, models can suffer on unseen data or target domain without such implicit feature, leading to unreliable performance. Second, in a real-world fake news detection system, a false

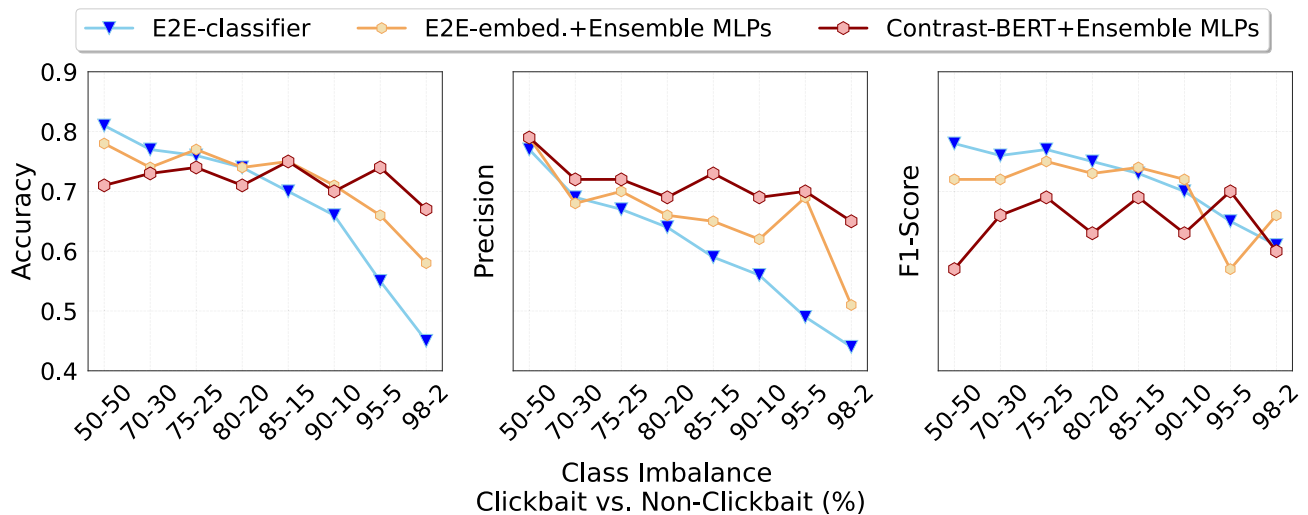


Fig. 6 Performance across different class imbalance ratio. Scores are computed based on “Clickbait” as positive class (class-id=1)

positive case where a Non-Hoax sample or legitimate news is incorrectly labeled as *Hoax* or *Fake*, can harm the reputation of the news source and the author of the reported claim [33].

We first annotate test data by utilizing OpenAI ChatGPT to generate binary labels (Neutral vs. Sensational) given short claims (Titles). We run LLM inference experiment three times to extract the label prediction based on the majority vote. The resulting annotated test data is composed of 182 samples with neutral tone and 412 samples with sensational tone. Both Hoax and Non-Hoax subsets contain balanced representation of samples with sensational tone (206 samples), inferring that sensational tone coexists in both classes. On the other hand, neutral tone is more skewed (43 Hoax samples vs. 139 Non-Hoax samples). Table 9 shows the scores representing bias measure for zero-shot classification task with BERT embedding models in this study. In general, Pairwise Contrast-BERT is less biased compared to the other models when predicting Hoax samples under both neutral and sensational tones.

Imbalance ratio

We investigate performance trend across different class imbalance ratio, ranging from balance class representation {50:50} to extremely imbalance ratio {98:2}, given binary class categories: clickbait versus nonclickbait. Based on results in Fig. 6, we observe that Contrast-BERT is more accurate when training data is extremely imbalanced (ratio $\geq 95:5$) based on accuracy and precision metrics. However, Contrast-BERT has lower F₁-score, inferring low recall score. This overall low recall score, similar to the real-world use case of fake news detection in Table 5, makes sense because Contrast-BERT does not explicitly learn explicit label supervision during training stage. Instead, the model exploits alternate training signals solely based on textual content. In contrast, E2E classifier learns skewed data distribution during training stage, thus, the supervised model carries bias that the majority samples in unseen target data are from the majority class category (Clickbait), resulting in higher recall yet lower precision.

6 Conclusion

In this study, we present a simple framework for low-resource fake news detection based on contrastive learning and stacking-based ensemble learning. Instead of viewing the problem as an end-to-end classification task, we decompose the fake news detection task into two main sequential stages: (1) learning useful embedding models for fake news detection on extremely imbalanced data; and (2) combining the resulting embedding models and

ensemble of MLPs as a classification pipeline. Our work demonstrates the capability of Contrast-BERT for addressing an extreme class imbalance problem for low-resource fake news detection tasks, particularly in Indonesian, Bangla, and Czech languages.

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Data and code availability Data, materials, and code implementation will be made publicly available. An anonymous temporary repository is available at <https://anonymous.4open.science/r/contrast-BERT>.

Declarations

Competing interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

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